

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	91002338
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture: Script-Controlled ACL on the ServiceNow platform

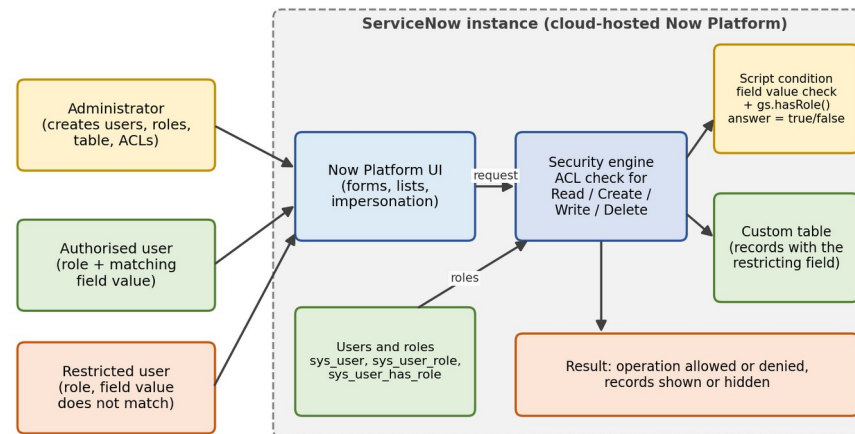


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
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1.	User Interface	Now Platform UI used by the administrator to build the setup, and by users to open forms and lists. Impersonation is used for testing.	ServiceNow Now Platform UI (forms, lists, impersonate user)
2.	Application Logic-1	Creates the users and roles and assigns roles to users (Milestone 1).	ServiceNow User Administration: sys_user, sys_user_role, sys_user_has_role
3.	Application Logic-2	Read and Create ACLs whose script checks the value of a field on the record and the user's role (Milestones 3 and 4).	ServiceNow ACL (sys_security_acl), Glide server-side JavaScript
4.	Application Logic-3	Write and Delete ACLs with the same field-value script logic (Milestones 5 and 6).	ServiceNow ACL (sys_security_acl), Glide server-side JavaScript
5.	Database	Custom table holding the records, with a field whose value controls who may access each record (Milestone 2).	ServiceNow relational database (managed by the platform)
6.	Cloud Database	Example: managed cloud database of the ServiceNow instance, holding the custom table and the ACL records. No separate database is set up by the team.	ServiceNow cloud instance database (managed by ServiceNow)
7.	File Storage	Example: supporting documents attached to a record (for instance a screenshot of a test result) are kept as record attachments.	ServiceNow attachments (sys_attachment), or Local Filesystem for exports
8.	External API-1	Example: REST call that lists the records a test user can see, used to check that the Read ACL also protects API access.	ServiceNow REST Table API

9.	External API-2	Example: single sign-on so that users sign in with their organisation account before ACLs are evaluated.	Microsoft Entra ID (Azure AD) SSO / SAML 2.0
10.	Machine Learning Model	Example: model that suggests the sensitivity value of a new record, so the restricting field is filled in consistently.	ServiceNow Predictive Intelligence (classification model)
11.	Infrastructure (Server / Cloud)	Application deployment on a ServiceNow instance hosted in the cloud by ServiceNow. Local system: any laptop with a web browser (e.g. Chrome) for administrators and testers Cloud server configuration: e.g. a ServiceNow Personal Developer Instance, managed by ServiceNow	ServiceNow cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is used. The solution is built with the platform's own scripting and configuration.	ServiceNow Glide API (JavaScript)
2.	Security Implementations	Role-based access plus scripted ACLs for Read, Create, Write and Delete. A record is accessible only when the user's role and the record's field value both allow it. Access is checked on the server, and every operation is tested by impersonating users.	ServiceNow roles, ACL scripts, impersonation
3.	Scalable Architecture	Rules are stored as ACL records, so the same pattern can be applied to more tables and roles	ServiceNow multi-tenant cloud platform

S.No	Characteristics	Description	Technology
		without changing the application.	
4.	Availability	Availability is provided by the ServiceNow cloud platform that hosts the instance.	ServiceNow cloud
5.	Performance	ACL scripts are kept short and only compare the current record's field value with the user's role, so they add little cost to each request.	Glide JavaScript in ACL conditions

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>